Noken:
implicit contextual token learning

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Abstract
Given a token in a context string, an implicit deep learning model reveals a latent token, that we dub a 'noken', and token representations, such that keys embue meaning to queries and values. This construction can then be construed as a machine oracle that aligns meaning utilizing a universal interlingua for next-token and sequence-to-sequence learning.

Noken – implicit contextual token learning:

$Tok = \{t_1, \dots, t_m\}$, token $t_i \in \mathbb{R}^r$

$Nok = \{s_1, \dots, s_m\}$, noken* $s_i \in \mathbb{R}^n$, $n > r$

Latent representations:

$$\begin{aligned} Tok &\longrightarrow Lat \subseteq \mathbb{R}^n \\ t_i &\longmapsto q_i = Q t_i, \quad Q \in \mathbb{R}^{n \times r} \end{aligned}$$

$$\begin{aligned} Tok &\longrightarrow Lat \\ t_i &\longmapsto {}^t k_i = {}^t K t_i, \quad {}^t K \in \mathbb{R}^{n \times r} \end{aligned}$$

$$\begin{aligned} Nok &\longrightarrow Lat \\ s_i &\longmapsto {}^s k_i = {}^s K s_i, \quad {}^s K \in \mathbb{R}^{n \times n} \end{aligned}$$

$$\begin{aligned} Nok &\longrightarrow Lat \\ s_i &\longmapsto v_i = V s_i, \quad V \in \mathbb{R}^{n \times n} \end{aligned}$$

Let $\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix} \in \mathbb{R}^n_+$, $\varepsilon_i \geq 0$, $i \in [n]$.

Given context $[t_1, \ldots, t_m] \in \mathbb{R}^{r \times m}$ and $\varepsilon \in \mathbb{R}^n$,

learn matrices $V, Q, {}^sK, {}^tK$ and fixed point

state $s_i \in \mathbb{R}^n$, from an implicit model so that:

$$s_i = \varphi(Vs_i + Qt_i)$$

$$\varepsilon = {}^sKs_i + (-{}^tK)t_i$$

where V is well-posed for activation function φ [1].

* This defines NoK and the latent representations.

We now apply this setting to next-token and

sequence-to-sequence learning.

Let score : Lat × Lat $\rightarrow \mathbb{R}$ be a similarity function,

like a reproducing kernel, for example.

Next-token learning

$$K$$

$$\underset{\in \mathbb{R}^n}{\text{token}}\ t_i \left\{ \begin{array}{l} {}^t k_i \\ \\ q_i \end{array} \right. \qquad \left. \begin{array}{r} {}^s k_i \\ \\ v_i \end{array} \right\} s_i\ \underset{\in \mathbb{R}^n}{\text{noken}}$$

$$Q \qquad\qquad\qquad\qquad V$$

Note that ${}^s k_i = {}^t k_i + \varepsilon$

<u>Input</u> target token t_i from context $[t_1, \ldots, t_m]$

Let $\alpha_{ij} \triangleq \underset{j \in [m]}{\text{softmax}}\ \text{score}(q_i, {}^s k_j)$

and $j_i^\alpha \triangleq \arg \max_{j \in [m]} \alpha_{ij}$.

Define $h_i \triangleq \sum_{j=1}^{m} \alpha_{ij} v_j \in \mathbb{R}^n$

Let $\beta_{ij} \triangleq \underset{j \in [m]}{\text{softmax}}\ \text{score}(h_i, {}^t k_j)$

and $j_i^\beta \triangleq \arg \max_{j \in [m]} \beta_{ij}$.

** Define $q_i' = q_{j_i^\beta}$

Define average Renyi α-entropy:

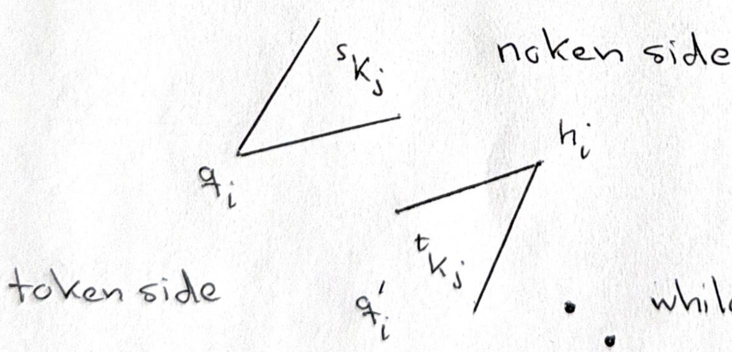
$$E(\{q_i\}_{i=1}^m) \triangleq \frac{1}{m} \sum_{i=1}^m H_\alpha^i$$

where $H_\alpha^i \triangleq H_\alpha(\{\beta_{ij}\}_{j=1}^m)$

If $E(\{q_i'\}_{i=1}^m) < E(\{q_i\}_{i=1}^m)$ (cf [2])

replace q_i with q_i' and continue updating q_i while average Renyi α-entropy declines or is small enough.

<u>Output</u> token $o_{i+1} \triangleq t_{j_i^\beta} \in \mathbb{R}^r$



$\therefore$ while E declines, update queries

In particular, note that $t_{j_i^\beta}$ depends on context, φ, V, $\mathcal{E}$ and score among other things. The implicit models connecting everything together.

So $[t_1, \ldots, t_m]$ yields $[o_2, \ldots, o_{m+1}]$.

The shift in indexing will become apparent.

<u>Token prediction:</u>

- Let t_1 = the empty token or a seed token
- From $[t_1]$, let $t_2 \triangleq o_2$
- From $[t_1, t_2]$, let $t_3 \triangleq o_3$
- From $[t_1, t_2, t_3]$, let $t_4 \triangleq o_4$

and so on, autoregressively generating the next token.

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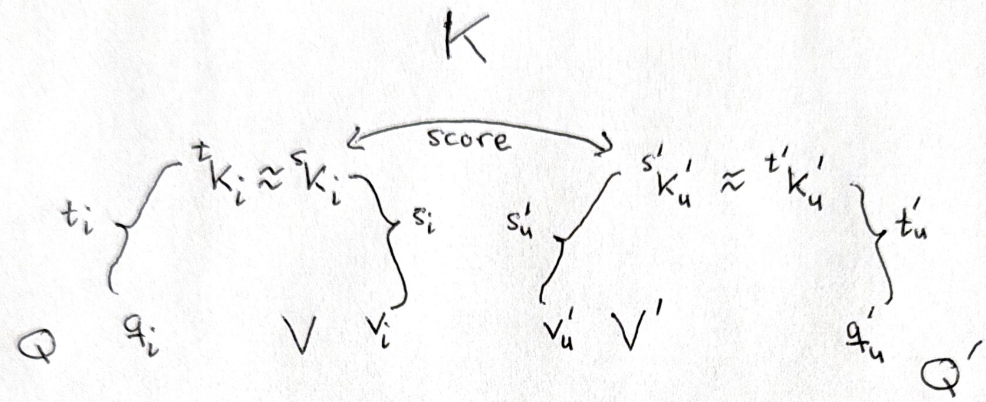
Sequence-to-sequence learning $[t_1, \ldots, t_m]$ to $[t'_1, \ldots, t'_w]$



$$s_i = \varphi(V s_i + Q t_i) \qquad s'_u = \psi(V' s'_u + Q' t'_u)$$

$$\varepsilon = {}^s K s_i + (-{}^t K) t_i \qquad \varepsilon' = {}^{s'}K' s'_u + (-{}^{t'}K') t'_u$$

Fix $i \in [m]$, let $\beta_{iu} = \underset{u \in [W]}{\mathrm{softmax}}\ \mathrm{score}({}^s k_i, {}^{s'}k'_u)$

and $u_i = \underset{u \in [W]}{\arg\max}\ \beta_{iu}$, where ${}^{s'}k_u = {}^{s'}K' s'_u$

(and $v'_u = V' s'_u$, ${}^{t'}k'_u = {}^{t'}K' t'_u$, $q'_u = Q' t'_u$)

So key ${}^{s'}k'_{u_i}$ is most similar to key ${}^s k_i$.

Let t'_{u_i} be the translation of t_i.

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2 Note that queries q_i and q'_u could be updated** if needed.

Discussion:

	keys K	
meaning	$[{}^tk_1,\dots,{}^tk_m]$	$[{}^sk_1,\dots,{}^sk_m]$
representation	$[q_1,\dots,q_m]$	$[v_1,\dots,v_m]$
	queries Q	values V
context	$[t_1,\dots,t_m]$	$[s_1,\dots,s_m]$
token	t_i	noken s_i

Think of keys as giving meaning to queries and values in that query q_i has features given by key ${}^tk_i$, likewise value v_i has features given by key ${}^sk_i$.

Suppose an agent $\mathbb{A}$ wants to predict t_{m+1} in the context $[t_1,\dots,t_m]$ or translate $[t_1,\dots,t_m]$ to t'_u for agent $\mathbb{B}$. Interpret implicit nokens s_i and s'_u as coming from a machine oracle $\mathbb{O}$.

In both tasks, after refining queries, $\mathbb{O}$ is using keys to align meaning. $\mathbb{A}$ speaks Q, $\mathbb{O}$ speaks V when talking to $\mathbb{A}$, and K is the universal language bridging the meaning of Q and the meaning of V. Similarly, $\mathbb{B}$ speaks Q' and $\mathbb{O}$ speaks V' when talking to $\mathbb{B}$. Now K acts as a universal interlingual translator from Q to Q'.

In summary, given a token t_i in context $[t_1, \ldots, t_m]$ an implicit model learns noken s_i, a latent token, as well as representations Q, K, V, so that keys embue meaning to queries and values. This construction can be construed as a machine oracle that utilizes a universal interlingua for next-token and sequence-to-sequence learning.

$\square$

References

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